

Thread 3 — Setting up Lean

AI Agents for Mathematics Research · setup and troubleshooting

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On a classroom machine

Lean, VS Code and the module repository are pre-installed and the Mathlib build cache is pre-warmed. You need only:

```
cd ~/thread3
git pull
code .
```

If `~/thread3` does not exist, or `git pull` fails, tell a demonstrator. Do not spend the session fixing it yourself; the image is our problem, not yours.

On your own machine

You need about **10 GB of free disk** and a working `git`.

1. Install `elan` (the Lean version manager). It reads the repository’s `lean-toolchain` file and installs exactly the right Lean version, so never install Lean any other way.

```
curl https://elan.lean-lang.org/elan-init.sh -sSf | sh -s -- -y
```

On Windows, use the installer from the Lean website rather than the shell script, or work inside WSL.

2. Install VS Code and, from the Extensions panel, the extension called **Lean4**, publisher `leanprover`. Do not install any other Lean extension.

3. Clone the repository and fetch the build cache.

```
git clone <module repo URL> thread3
cd thread3
lake exe cache get
```

`lake exe cache get` downloads Mathlib pre-built. It takes a few minutes and several gigabytes. **Do not skip it** — without it, Lean will build Mathlib from source, which takes hours.

4. Open the folder in VS Code — `code .` from inside `thread3`, or File ► Open Folder. Opening a single `.lean` file instead of the folder is the single commonest setup mistake: Lean then cannot find the project and nothing works.

5. Check it. Open `Thread3/Week01.lean`. Within a minute or so the Infoview should appear on the right showing goals. If it does, you are done.

Using it

Infoview	opens automatically; reopen with Ctrl/Cmd+Shift+Enter
Goals	shown in the Infoview for wherever your cursor is
Errors	red squiggles, and the Problems panel (Ctrl/Cmd+Shift+M)
Unicode	type <code>\to</code> , <code>\forall</code> , <code>\R</code> , <code>\N</code> , <code>\le</code> , <code>\in</code> , <code>\sum</code> , <code>\<</code> and press space or tab
Restart a file	Ctrl/Cmd+Shift+P ► “Lean 4: Restart File”

A `sorry` produces a **warning**, not an error, and the file still builds. That is why “it builds” is not the standard for finished work — no `sorry` and an empty Problems panel is.

Troubleshooting

The Infoview never appears, or says “Waiting for Lean server”. You opened a file rather than the folder. Close everything, open the `thread3` folder, and try again.

Everything is underlined red, including `import`. Mathlib is not built. Run `lake exe cache get` in the project root, then restart VS Code.

It is building Mathlib from source (thousands of files scrolling past). Stop it. `lake exe cache get` failed — usually a network problem or a mismatched Lean version. Check that `elan show` reports the version in `lean-toolchain`.

lake: command not found. `elan` is not on your `PATH`. Open a new terminal; if that does not fix it, on macOS or Linux add `source $HOME/.elan/env` to your shell profile.

A lemma name that works for a classmate fails for you. You are on a different Mathlib revision. Run `git pull` and `lake exe cache get`. Mathlib renames things, and the module pins one revision for exactly this reason — never edit `lean-toolchain` or `lakefile.toml`.

Everything worked yesterday and now nothing does. Try “Lean 4: Restart File”, then restart VS Code, then `lake exe cache get` again, in that order.

It is very slow. Lean recompiles the whole file on every keystroke-idle. Large `simp` calls, `exact?` and `decide` are genuinely slow; that is normal. If the whole file is slow, check you have the cache and not a source build.

Getting help

In the session, ask a demonstrator — that is what they are there for, and being stuck for ten minutes on a setup problem is a waste of the fifty. Outside the session, post the **exact error text** in the module forum, not a screenshot and not a description. Include the output of `elan show` and the first line of `git log --oneline -1`. Anyone can then reproduce your problem in a minute.